

n8n Automation Workflow

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Course	TechMa Zone n8n Automation
Tool Used	n8n Cloud (n8n.io)

01 What is This Assignment About?

This assignment shows how to build an automated workflow using n8n — a free automation tool. When someone fills out a form, n8n automatically saves their data to Google Sheets and sends them a custom email based on their occupation. No manual work needed!

02 Full Workflow Block Diagram

Below is the complete n8n workflow. Each box (called a node) does one specific task. They are connected in order so data flows automatically from start to finish.

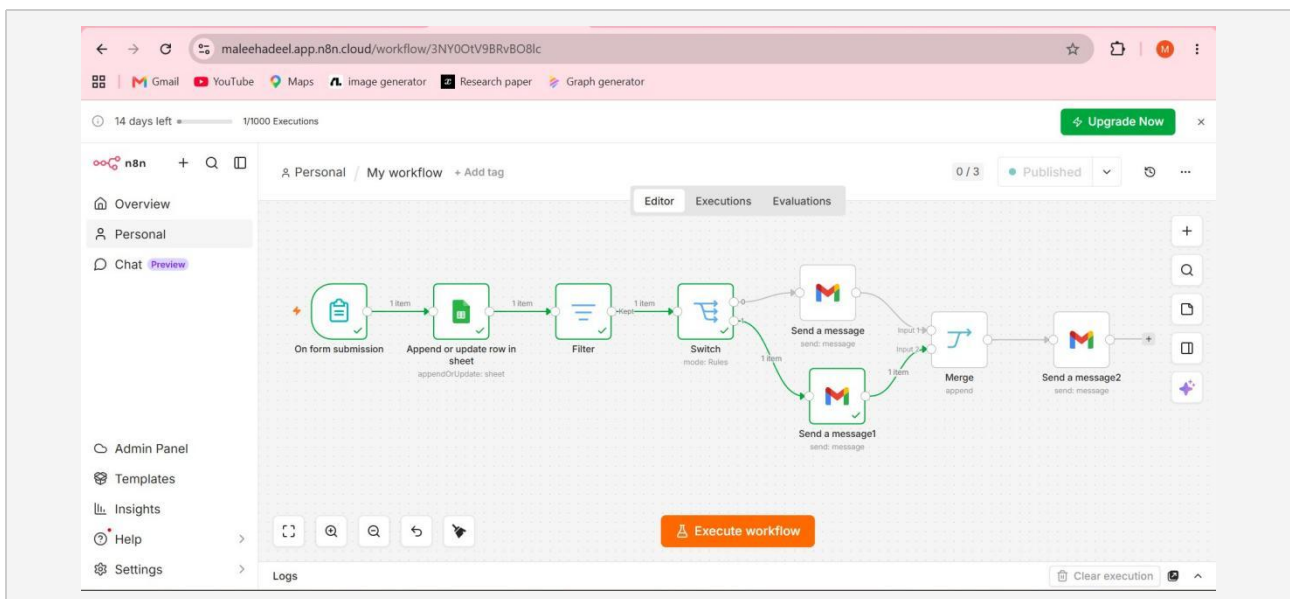


Figure 1: Complete n8n Workflow on Canvas

Step	Node	What it does
1	On Form Submission	Collects data when someone fills the form
2	Append or Update Row in Sheet	Saves data to Google Sheets (no duplicates)
3	Filter	Blocks Students from going further
4	Switch	Sends Doctors and Engineers on separate paths
5	Send a Message (Doctor)	Sends a special email to Doctors
5	Send a Message 1 (Engineer)	Sends a special email to Engineers
6	Merge	Joins both paths back together
7	Send a Message 2	Sends one final email to everyone who passed

03 Step 1: Form Node : Collecting Data

A form was created with 5 fields. People fill this form and their data is sent to n8n automatically.

The screenshot shows the 'Parameters' tab of the n8n Form Node configuration. At the top, there are tabs for 'Parameters' and 'Settings', and an orange button labeled 'Execute step' with a flask icon. Below the tabs, the 'Form Elements' section is visible. It contains two form fields. The first field has a 'Label' of 'First Name' and an 'Element Type' of 'Text Input'. Below the first field is a grey button labeled 'Add Attributes'. The second field has a 'Label' of 'Last Name' and an 'Element Type' of 'Text Input'. A vertical scrollbar is on the right side of the form elements list.

Figure 2: Form Fields : First Name and Last Name

The screenshot shows the 'Parameters' tab of the n8n Form Node configuration, continuing from the previous one. It shows the last two form fields. The third field has a 'Label' of 'Email' and an 'Element Type' of 'Email'. Below it is a grey button labeled 'Add Attributes'. The fourth field has a 'Label' of 'Phone #' and an 'Element Type' of 'Text Input'. A vertical scrollbar is on the right side of the form elements list.

Figure 3: Form Fields : Email and Phone Number

Parameters Settings Execute step

Element Type
Number

Add Attributes

Label
Occupation

Element Type
Dropdown

Field Options

Option
Student

Figure 4: Occupation Dropdown : Student option

Parameters Settings Execute step

Option
Teacher

Option
Doctor

Option
Engineer

Add Field Option

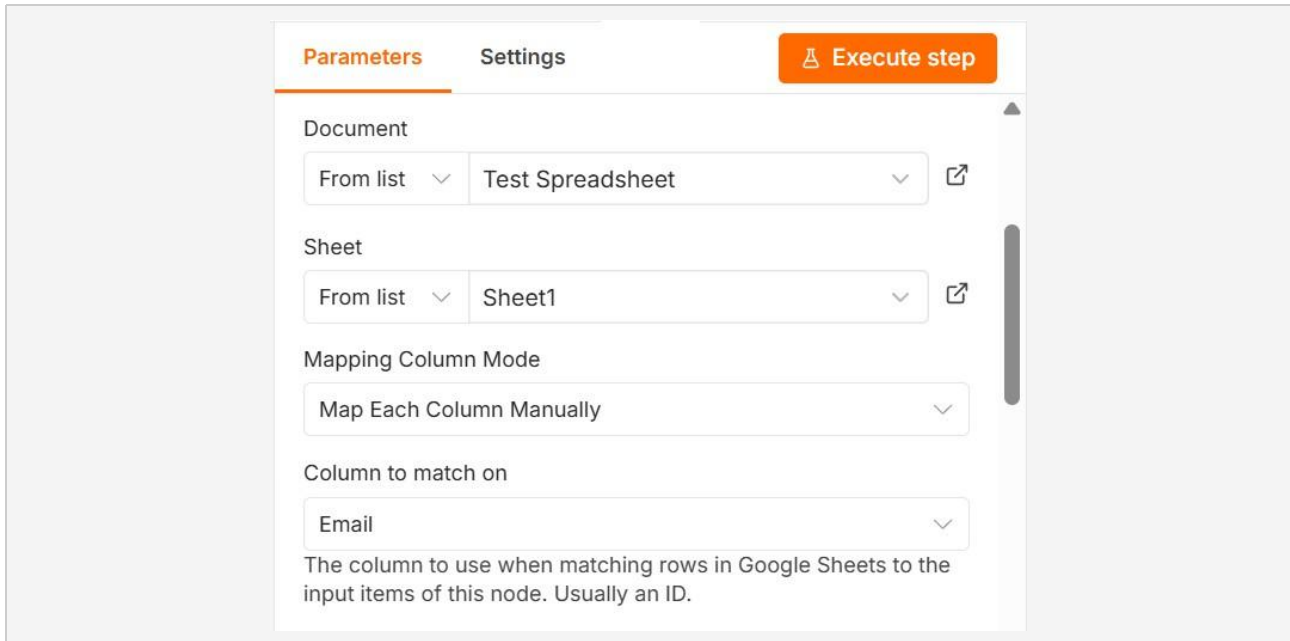
Add Attributes

Figure 5: Occupation Dropdown : Teacher, Doctor, Engineer

Field	Type
First Name	Text Input
Last Name	Text Input
Email	Email Input
Phone #	Number Input
Occupation	Dropdown (Student, Teacher, Doctor, Engineer)

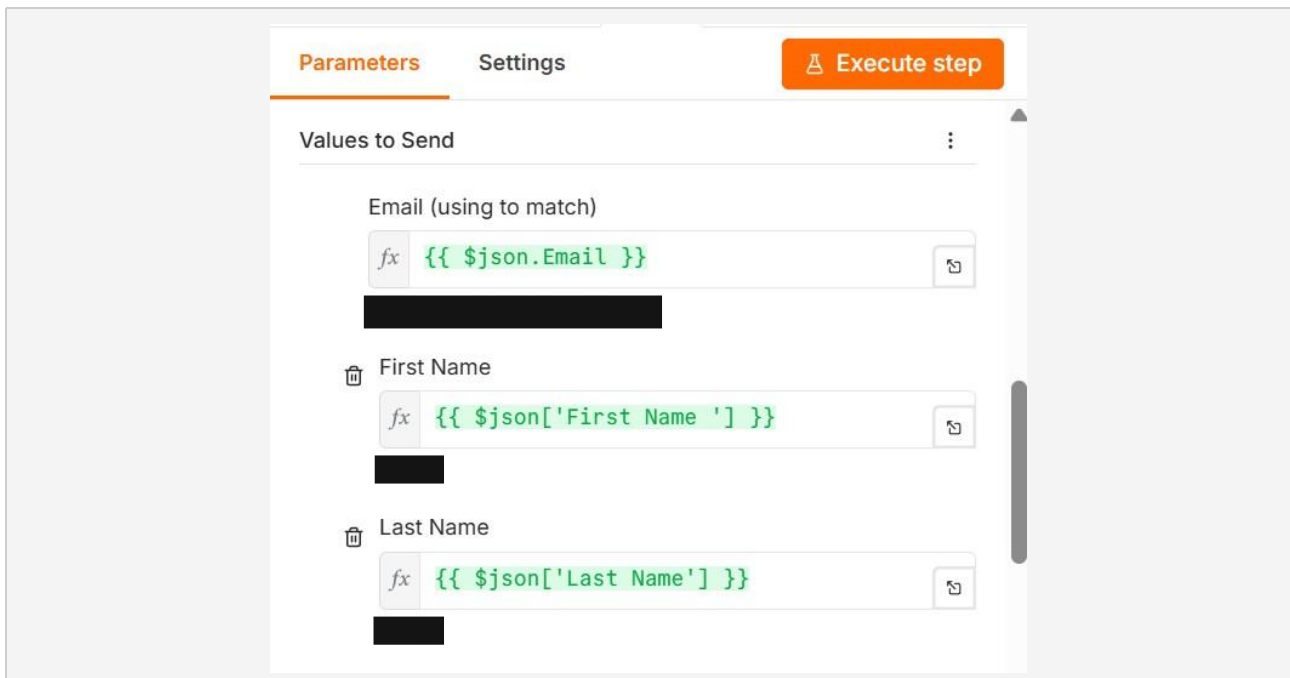
04 Step 2: Google Sheets : Saving Data

Every form submission is saved automatically to a Google Sheet called 'Test Spreadsheet'. The node uses 'Append or Update' mode this means if the same email submits again, it updates the existing row instead of adding a duplicate.



The screenshot shows the 'Parameters' tab of a node configuration. At the top right is an orange button labeled 'Execute step' with a flask icon. Below the tabs, there are four sections: 'Document' with a 'From list' dropdown and a text field containing 'Test Spreadsheet'; 'Sheet' with a 'From list' dropdown and a text field containing 'Sheet1'; 'Mapping Column Mode' with a dropdown set to 'Map Each Column Manually'; and 'Column to match on' with a dropdown set to 'Email'. A descriptive text at the bottom states: 'The column to use when matching rows in Google Sheets to the input items of this node. Usually an ID.'

Figure 6: Google Sheets Node : Document and Sheet Selection



The screenshot shows the 'Parameters' tab of the same node configuration. The 'Values to Send' section is expanded, showing three rows. Each row has a trash icon on the left, a label, and a text field with a formula. The first row is labeled 'Email (using to match)' and contains the formula `{{ $json.Email }}`. The second row is labeled 'First Name' and contains the formula `{{ $json['First Name'] }}`. The third row is labeled 'Last Name' and contains the formula `{{ $json['Last Name'] }}`. Each text field has a small icon on the right.

Figure 7: Column Mapping : Email, First Name, Last Name

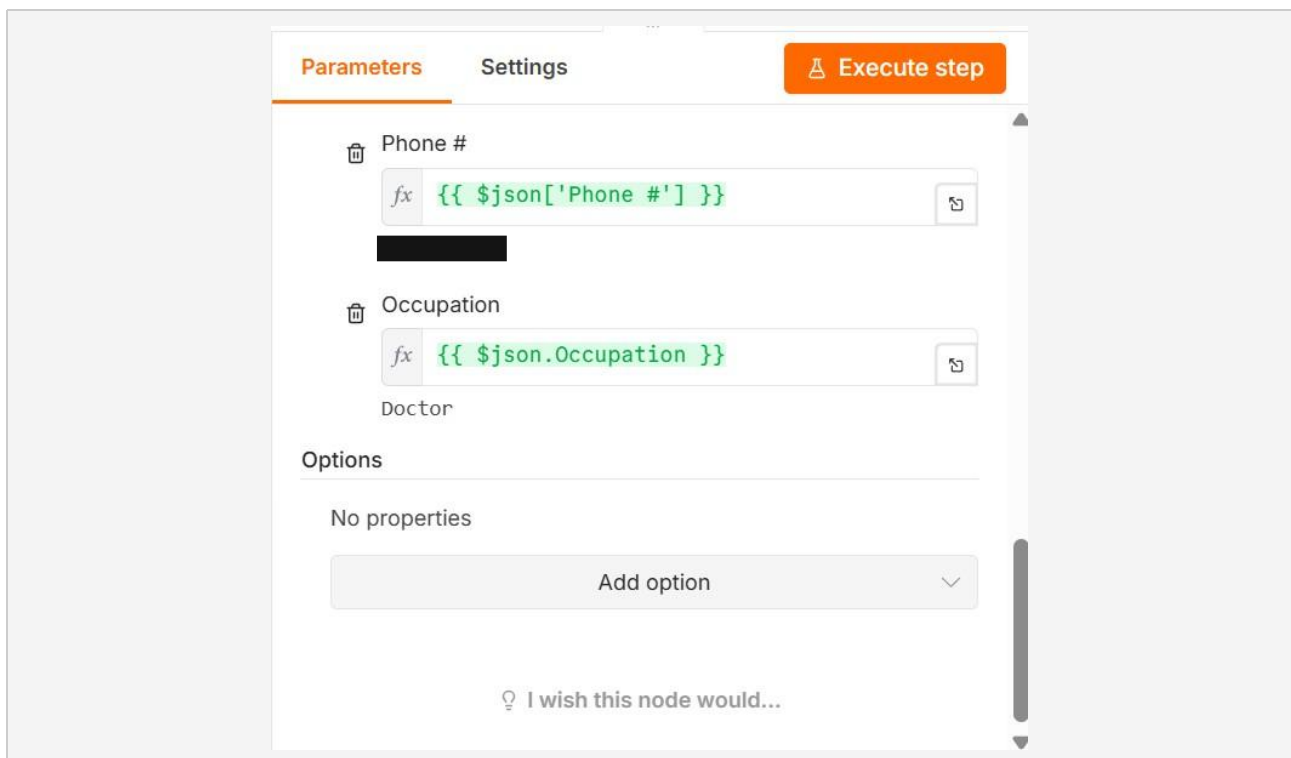


Figure 8: Column Mapping : Phone and Occupation

05 Step 3: Filter Node : Removing Students

The Filter node checks the Occupation field. If the person selected 'Student', they are blocked and no email is sent to them. Everyone else (Teacher, Doctor, Engineer) passes through.

The screenshot shows the configuration interface for a Filter Node. It has two tabs: 'Parameters' (selected) and 'Settings'. In the top right corner, there is an orange button labeled 'Execute step' with a play icon. The 'Parameters' tab contains a 'Conditions' section with a text input field containing the JSON expression `{{ $json.Occupation }}`, followed by a dropdown menu set to 'T is not equal to' and a checkmark icon. Below this, the value 'Student' is entered in a text field. A '+ Add condition' button is located below the condition field. Underneath, there is a toggle switch for 'Convert types where required', which is currently turned off. The 'Options' section below shows 'No properties' and an 'Add option' button with a dropdown arrow. A vertical scrollbar is visible on the right side of the configuration area.

Figure 9: Filter Node : Block Students

Occupation	Filter Result	Gets Email?
Student	BLOCKED	No
Teacher	PASSES	No (no Switch route)
Doctor	PASSES	Yes
Engineer	PASSES	Yes

06 Step 4: Switch Node : Splitting by Occupation

The Switch node reads the Occupation and sends the data on two different paths. Engineers go to one Gmail node and Doctors go to another. This way each group gets a different email.

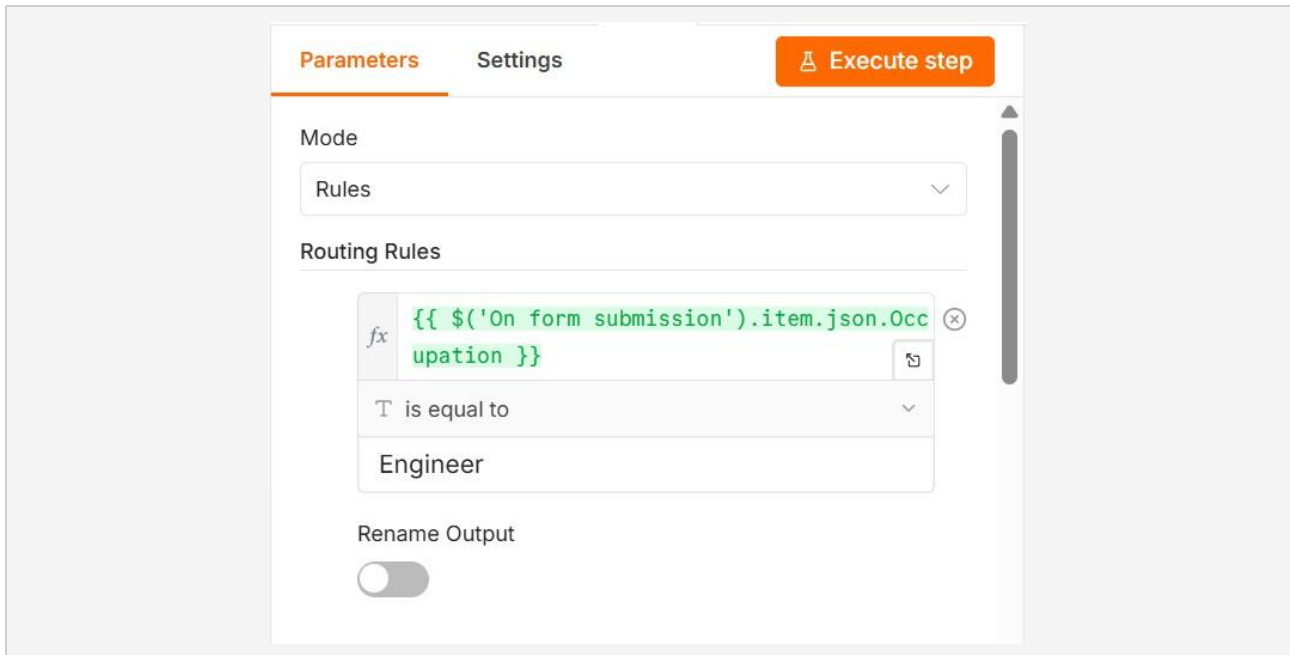


Figure 10: Switch Rule : Engineer Route

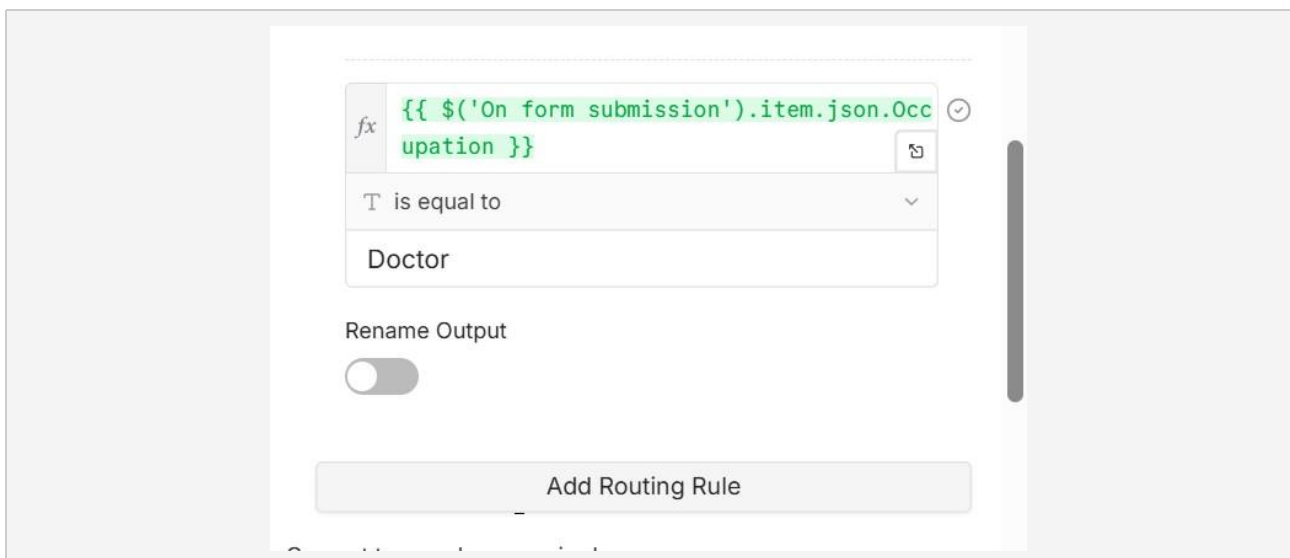
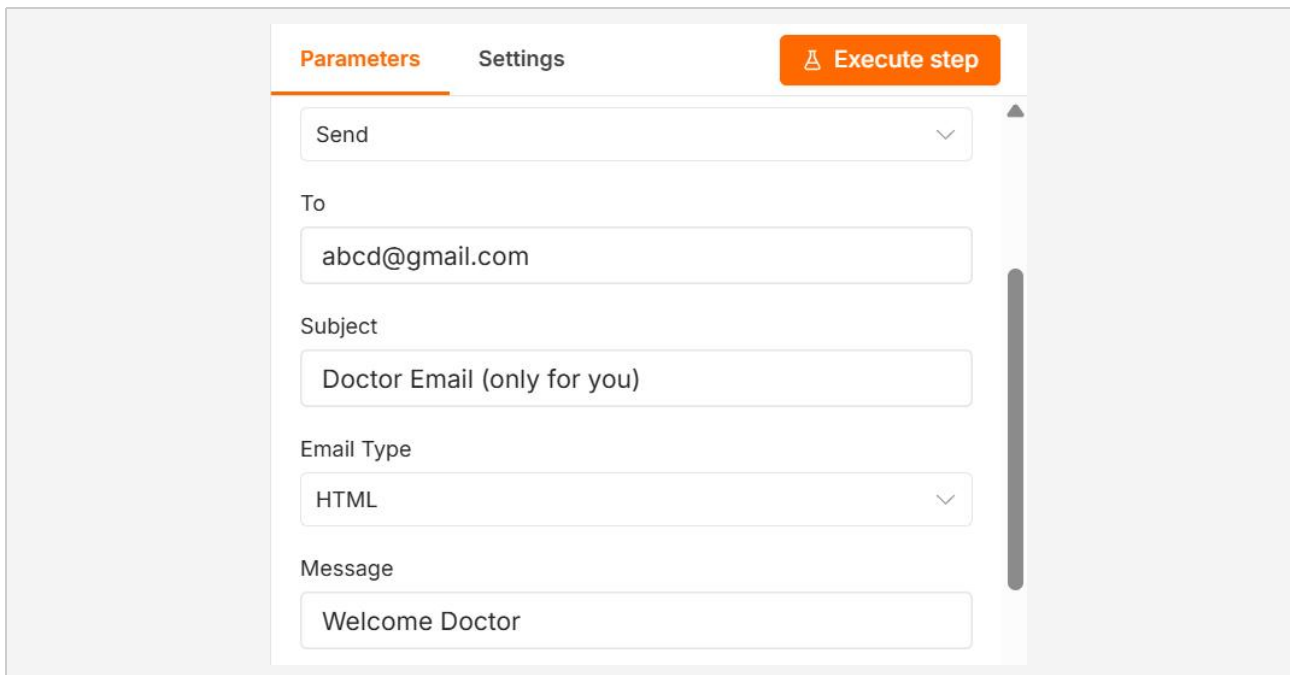


Figure 11: Switch Rule : Doctor Route

07 Step 5: Gmail Nodes : Sending Emails

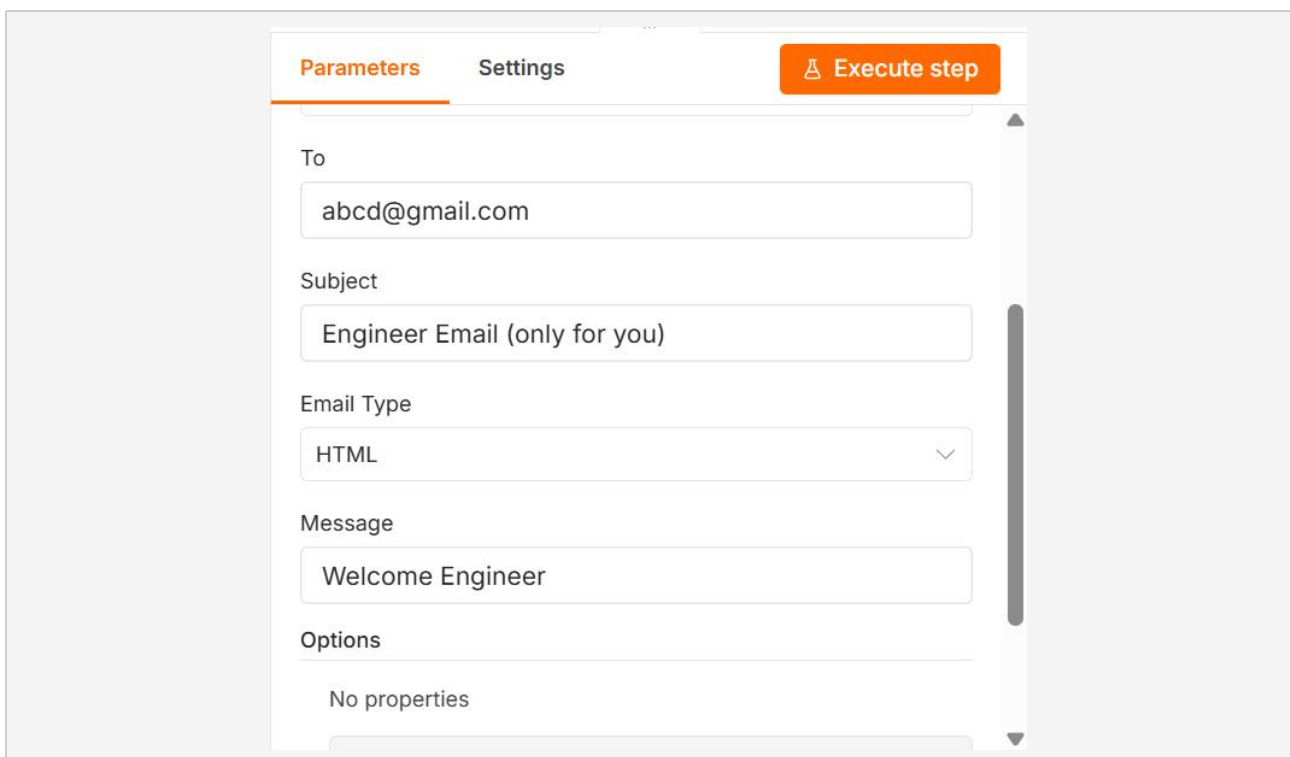
Two separate Gmail nodes send different emails. One is for Doctors and one is for Engineers. Each email has a different subject and message.



The screenshot shows the 'Parameters' tab of a Gmail node configuration. At the top, there are two tabs: 'Parameters' (selected) and 'Settings'. To the right of the tabs is an orange button with a flask icon and the text 'Execute step'. Below the tabs, the configuration fields are as follows:

- Send:** A dropdown menu with 'Send' selected and a downward arrow.
- To:** A text input field containing 'abcd@gmail.com'.
- Subject:** A text input field containing 'Doctor Email (only for you)'.
- Email Type:** A dropdown menu with 'HTML' selected and a downward arrow.
- Message:** A text input field containing 'Welcome Doctor'.

Figure 12: Gmail Node : Doctor Email Settings



The screenshot shows the 'Parameters' tab of a Gmail node configuration for an engineer. At the top, there are two tabs: 'Parameters' (selected) and 'Settings'. To the right of the tabs is an orange button with a flask icon and the text 'Execute step'. Below the tabs, the configuration fields are as follows:

- To:** A text input field containing 'abcd@gmail.com'.
- Subject:** A text input field containing 'Engineer Email (only for you)'.
- Email Type:** A dropdown menu with 'HTML' selected and a downward arrow.
- Message:** A text input field containing 'Welcome Engineer'.
- Options:** A section header with a text input field below it containing 'No properties'.

Figure 13: Gmail Node : Engineer Email Settings

Parameters

Settings

 Execute step

To

efgh@gmail.com

Subject

Mandatory Email

Email Type

HTML

Message

Hye hope you are good

Options

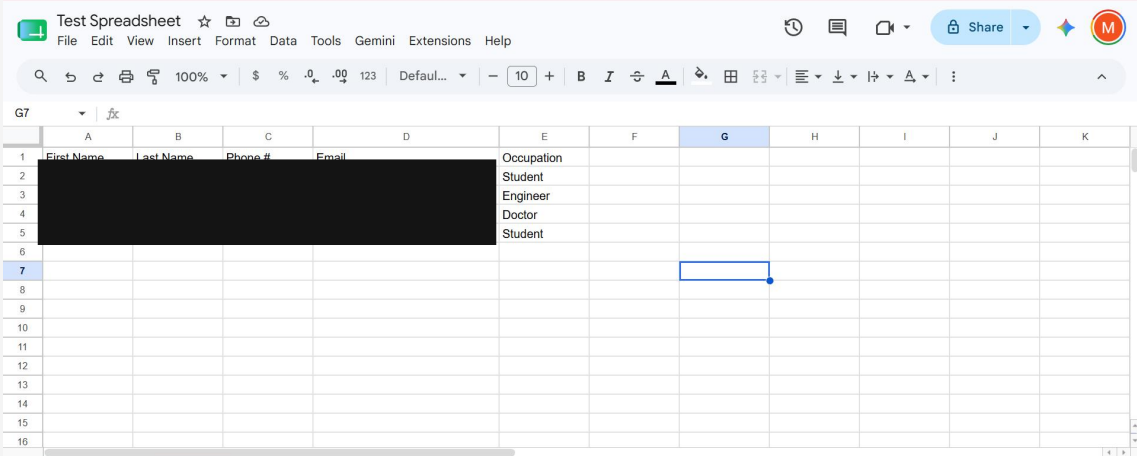
No properties

Add option

Figure 14: Gmail Node : After Merge Email Settings

8.1 Google Sheet : Collected Data

All form submissions were saved correctly to the Google Sheet. Four people submitted the form with different occupations.



	A	B	C	D	E	F	G	H	I	J	K
1	First Name	Last Name	Phone #	Email	Occupation						
2					Student						
3					Engineer						
4					Doctor						
5					Student						
6											
7											
8											
9											
10											
11											
12											
13											
14											
15											
16											

Figure 15: Google Sheet with Collected Form Data

8.2 Email Received : Doctor

When a Doctor filled the form, a 'Doctor Email (only for you)' email was sent automatically with the message 'Welcome Doctor'.

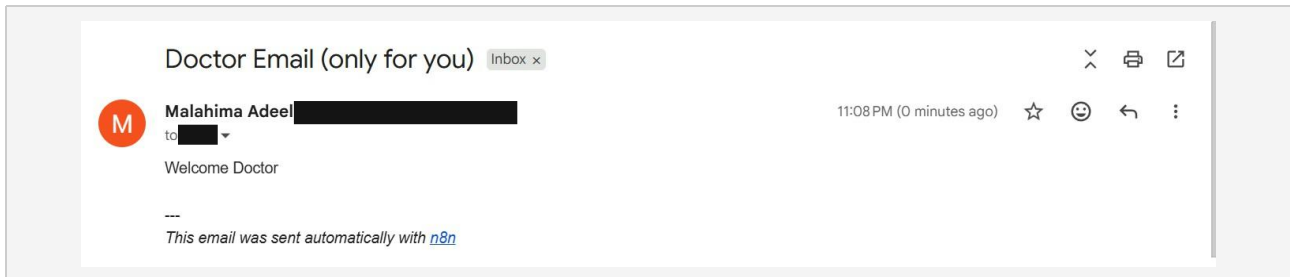


Figure 16: Email Received by Doctor

8.3 Email Received : Doctor

When an Doctor filled the form, a 'Hye hope you are good' email was sent automatically after merge.

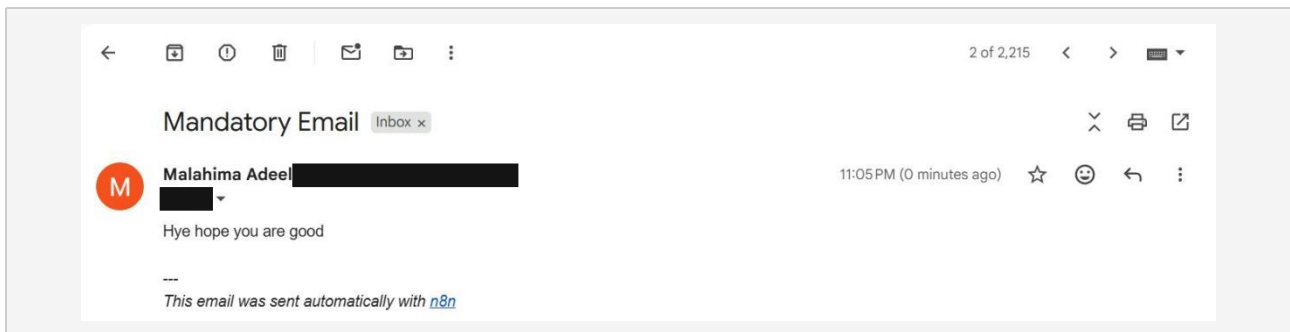


Figure 17: Email Received by Doctor after merging

09 Public Form Link

The form is live and can be shared with anyone. Click the link below to open it:



10 Conclusion

This assignment showed how n8n can automate a full data collection and email sending process without writing any code. The workflow successfully:

- Collected form data from multiple users
- Saved all responses to Google Sheets automatically
- Filtered out Students so they don't receive emails
- Split Doctors and Engineers onto separate email paths
- Sent different welcome emails to each occupation group
- Merged both paths and sent a final common email